SUPPORTING ADULT GUIDANCE

Angles in the same segment

Circle Theorems · Lesson 4. For any supporting adult — teacher, practitioner, parent or carer.
You do not need to be a mathematics specialist to support this lesson.

What this lesson does

This lesson establishes that angles subtended by the same arc (in the same segment) are equal, as a second corollary of the angle-at-the-centre theorem. Learners discover the invariance by dragging two apex points along a common arc in the Circle Theorem Lab and comparing the two angles, then prove the result — each inscribed angle is half the same central angle — and use it, with reasons, to find missing angles. It also sets up the contrast with the cyclic-quadrilateral case (points on opposite sides give supplementary angles).

Driving Question: *Why don't these angles change as the point slides round?*

The mathematics, in plain terms

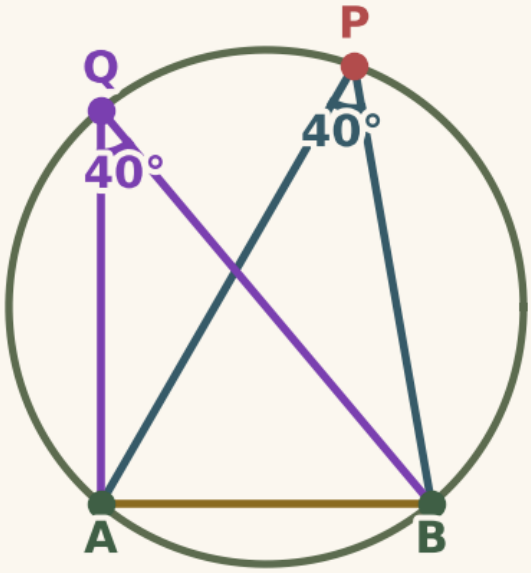
For a fixed chord AB, all inscribed angles subtending AB from the same arc are equal, because each equals half the central angle ∠AOB subtended by that arc. Points on the opposite arc subtend the supplementary angle (leading to the cyclic-quadrilateral result). The theorem is used to find equal angles in circle figures and, combined with the triangle angle sum, to solve multi-step angle problems.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Equal angles in the same segment

P and Q are on the same arc of a circle. ∠APB = 40°. Find ∠AQB.

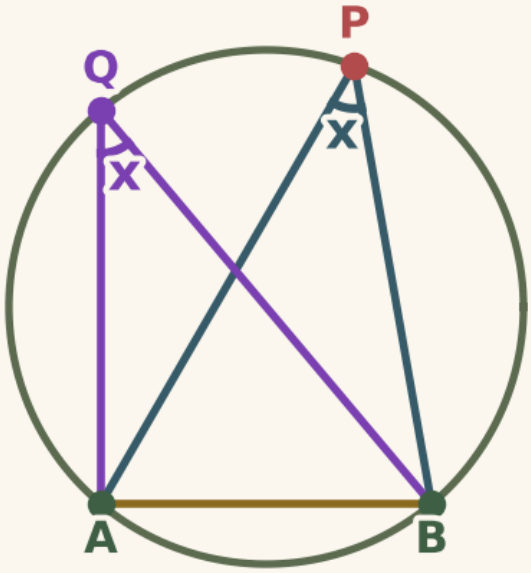


1. P and Q stand on the same arc AB (same side of the chord).	same segment
2. Angles in the same segment are equal.	∠AQB = ∠APB

∠AQB = 40°, because angles in the same segment are equal (each is half the same central angle).

Example 2 — Why they are equal

Prove ∠APB = ∠AQB for P and Q on the same arc. Let the angle at the centre be ∠AOB = 2x.



1. Draw the angle at the centre ∠AOB, and call it 2x.	∠AOB = 2x
2. By the keystone, ∠APB = ½ × 2x = x.	∠APB = x
3. ∠AQB is half the SAME central angle, so ∠AQB = x too.	∠AQB = x

∠APB = ∠AQB = x. Both are half the one central angle 2x, so they are equal — angles in the same segment. ■

See them stay equal, then prove it once

Let the learner drag both apex points and watch the two angles track each other before stating the rule. Frame the proof, as with the semicircle, as the keystone applied twice to the same central angle — reinforcing rather than adding to what must be remembered. Insist on the reason 'angles in the same segment' with every use, and make the same-side / opposite-side distinction explicit, since it both prevents errors here and previews the cyclic-quadrilateral theorem next lesson.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing

Connection	See two points on one arc give the same angle.	~9 min
Movement	Find equal angles in the same segment, with reasons.	~11 min
Reflection	Circle Theorem Lab: drag two apexes, the angles stay equal.	~12 min
Creativity	Prove it: both are half the same centre angle.	~9 min
Rest	A short pause on one idea seen from three positions.	~2 min
Nutrition	Intellectual: getting a new theorem from one you have.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Assuming any two inscribed angles are equal.
'Do both angles stand on the SAME chord, from the SAME arc? Only then are they in the same segment and equal.'

Giving the value with no reason.
'Correct — and the reason is "angles in the same segment". Name it every time.'

Confusing same segment with opposite segments.
'Are the two points on the same side of the chord, or opposite sides? Same side: equal. Opposite: they add to 180°.'

Not spotting the shared chord in a busy figure.
'Which chord do both angles open onto? Trace it — the two angles must share the same chord and arc.'

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: the angle at the centre theorem (Lesson 2), the idea of an angle standing on an arc, and what a segment of a circle is (Lesson 1). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner cannot see why two angles are equal, or gives a value with no justification. The aim is a question that surfaces the shared chord and the reason.

- "Do these two angles stand on the same chord, from the same arc?"

- “If they are in the same segment, how do their sizes compare?”
- “What is your reason? Name the theorem.”
- “Are the points on the same side of the chord, or opposite sides?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Why are the angles equal?

Because each one is half the same central angle ∠AOB. The chord AB fixes that central angle, so every inscribed angle on the same arc is the same half of it.

How is this different from the angle at the centre?

It is the same theorem used twice. Instead of comparing one circumference angle to the centre, we compare two circumference angles to each other — both equal to half the same centre angle.

What if the points are on opposite sides of the chord?

Then they are in different segments and the angles are not equal — they are supplementary (add to 180°). That is the cyclic-quadrilateral theorem, coming next.

How do I spot it in an exam?

Look for two angles at the circumference opening onto the same chord from the same side. That pattern — often drawn as a 'bowtie' or 'arrowhead' — signals angles in the same segment.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.

Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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